UNIT - I

Engineering Seismology & Theory of Vibrations

Syllabus (as given):

Earthquake phenomenon - causes - faults - plate tectonics - seismic waves - terms - magnitude/intensity - scales - energy - instruments (seismogram, seismoscope, seismograph) - strong ground motion - seismic zones of India. Elements of vibratory system - DOF - continuous system - lumped mass - SHM - free vibration of SDOF (undamped/damped) - critical damping - log decrement - forced vibration - DMF - rigid base excitation - earthquake ground motion.

PART A : ENGINEERING SEISMOLOGY

1.1 Earthquake Phenomenon

Definition - Earthquake

A sudden release of strain energy stored in the earth's crust/upper mantle, which travels as seismic waves and shakes the ground surface.

- *Earth's structure: Crust (5-70 km) -> Mantle (~2900 km thick) -> Outer core (liquid, to ~5150 km) -> Inner core (solid, radius ~1220 km).*
- *Lithosphere (crust + top mantle, ~100 km) is rigid and broken into plates; below it the soft asthenosphere flows slowly (convection currents).*
- *Strain builds up for years at plate edges/faults; when rock strength is exceeded it ruptures and slips -> earthquake.*
- *Most earthquakes (~90%) are tectonic and occur at depths < 70 km (shallow), hence very destructive.*

Term	Meaning
Focus / Hypocentre	Point inside earth where rupture starts and energy is released
Epicentre	Point on ground surface vertically above the focus
Focal depth	Vertical distance focus -> epicentre (shallow < 70 km, intermediate 70-300 km, deep > 300 km)
Epicentral distance	Distance from the observation station to epicentre
Hypocentral distance	Distance from station to focus
Foreshock / Aftershock	Smaller shocks before / after the main shock in the same region
Isoseismals	Lines joining places of equal intensity on a map
Meizoseismal area	Area of maximum damage, surrounding the epicentre
Seismic gap	Active fault segment that has not ruptured for long: likely site of future shock

Terms associated with earthquakes

1.2 Causes of Earthquakes

- Tectonic (most important): sudden slip along faults / plate boundaries - Elastic Rebound Theory (Reid, 1906, after the 1906 San Francisco EQ).
- Volcanic: movement of magma and gases; usually small, local.
- Induced / man-made: reservoir-induced seismicity (Koyna 1967, M 6.5), mining, deep well injection, nuclear blasts.
- Others: landslides, rock-falls, meteorite impact, collapse of caverns.

Important / Exam point

Elastic Rebound Theory: rocks on either side of a fault are slowly strained -> strain exceeds frictional strength -> sudden slip -> rocks "rebound" to a less-strained position and the stored elastic energy is released as waves.

1.3 Faults

Definition - Fault

A fracture/zone in the rock across which there has been relative displacement. Dip = angle of fault plane from horizontal; Strike = direction of fault line on the surface.

Type	Movement	Stress / Notes
Normal (gravity) fault	Hanging wall moves DOWN relative to footwall	Tension; crust extends (divergent zones)
Reverse / Thrust fault	Hanging wall moves UP over footwall	Compression; crust shortens (Himalaya). Thrust = low dip (<45°)
Strike-slip fault	Horizontal sliding, along the strike	Shear; right-lateral or left-lateral (San Andreas)
Oblique-slip fault	Combination of dip-slip and strike-slip	Most real faults

1.4 Plate Tectonics

Earth's lithosphere is divided into 7 major plates (Pacific, Eurasian, African, North American, South American, Indo-Australian, Antarctic) and several minor ones. Plates move 2-15 cm/year driven by mantle convection, ridge-push and slab-pull.

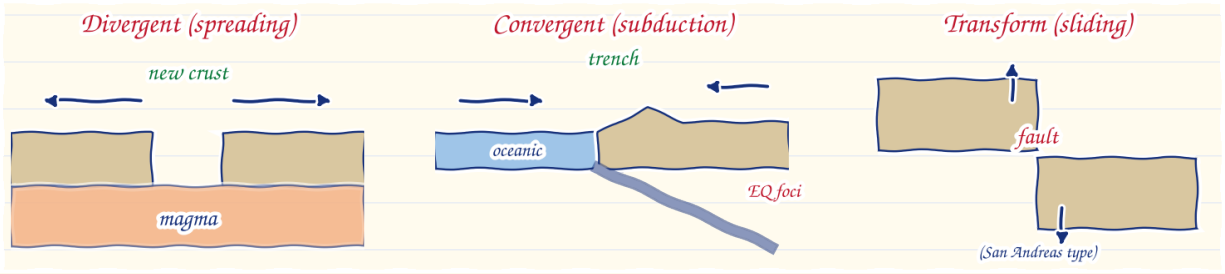


Fig: Types of plate boundaries

- Divergent boundary - plates move apart; new crust forms (Mid-Atlantic ridge); shallow, small EQs.
- Convergent boundary - plates collide: ocean-continent (subduction, deep EQs, volcano arcs - Andes), ocean-ocean (island arcs - Japan), continent-continent (mountain building - Himalaya:

Indian plate moves ~5 cm/yr north under Eurasia).

- *Transform boundary* - plates slide past; no crust created or destroyed; shallow strong EQs (San Andreas).
- *Intra-plate earthquakes* occur inside plates at old weak zones, e.g. Latur 1993, Bhuj 2001, Koyna - India's peninsular region.

> *Important / Exam point*
> India: the Indian plate collides with Eurasia -> Himalayan belt is among the most seismically active zones; 1897 Shillong, 1905 Kangra, 1934 Bihar-Nepal, 1950 Assam, 2001 Bhuj, 2015 Nepal.

1.5 Seismic Waves

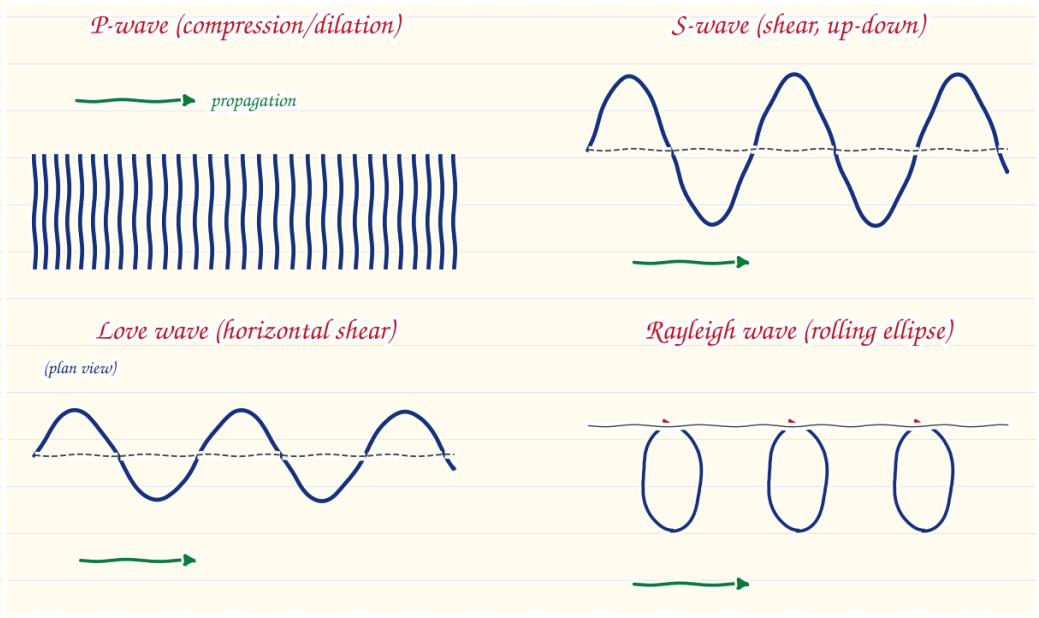


Fig: Seismic wave types

Wave	Class	Motion	Speed / Remarks
P (Primary)	Body, fastest	Particle motion parallel to travel (push-pull, compression)	$V_p = \sqrt{[(K+4\mu/3)/\rho]}$ ~ 5-8 km/s; travels through solid and liquid; first arrival
S (Secondary)	Body	Perpendicular to travel (shear)	$V_s = \sqrt{(\mu/\rho)}$ ~ 3-4.5 km/s; only solids; arrives second; causes most structural damage
Love (L)	Surface	Horizontal shear, perpendicular to travel	Slightly slower than S; very damaging to foundations
Rayleigh (R)	Surface	Retrograde elliptical in vertical plane (like ocean waves)	Slowest (~0.9 V_s); large amplitude; long period

- Order of arrival: P -> S -> Love -> Rayleigh. S-P time lag gives epicentral distance (about 8 km per second of lag).
- Surface waves decay slowly with distance, body waves faster. Surface waves dominate far-field and long-period motion (tall buildings, bridges).

- S-waves not passing through the outer core proves that the outer core is liquid (shadow zone).

1.6 Magnitude and Intensity

Definition - Magnitude (M)

A quantitative, instrumental measure of the energy released at the source. One value per earthquake, independent of location.

Richter local magnitude (1935)

$M_L = \log_{10} A - \log_{10} A_0(\delta)$

A = max trace amplitude (mm) on a Wood-Anderson seismograph at 100 km; A_0 = standard amplitude.

Other scales: body-wave m_b, surface-wave M_s, moment magnitude $M_w = (2/3) \log_{10} M_0 - 10.7$ ($M_0 = \mu A D$ in dyne-cm; does not saturate - preferred today).

Energy released (Gutenberg-Richter)

$\log_{10} E = 1.5 M + 4.8$ (E in joules)

Each unit increase in M -> amplitude x10, energy x 31.6 (~32). Two units -> energy x 1000.

e.g. M 7 releases 32 times the energy of M 6.

Definition - Intensity (I)

A qualitative measure of effects/damage at a given place (felt by people, damage to structures, ground changes). Differs from place to place; highest near epicentre.

Scale	Range	Remarks
Modified Mercalli (MMI)	I - XII	Most used internationally: I = not felt ... VII = damage to poorly built structures ... XII = total destruction
Rossi-Forel	I - X	Older scale
MSK-64 (Medvedev-Sponheuer-Karnik)	I - XII	Used in India (IS 1893 uses MSK); defines building types A, B, C
EMS-98 (European Macroseismic)	I - XII	Modern, vulnerability-class based
Richter / Moment	0 - ~9.5 (open ended)	Magnitude scales, log based

Magnitude vs Intensity	
Magnitude	Intensity
Energy at source; single number	Effect at a location; varies with distance, soil, building quality
Measured by seismograph	Assessed from observations / questionnaires
Arabic numerals (6.5)	Roman numerals (VII)

1.7 Earthquake Measuring Instruments

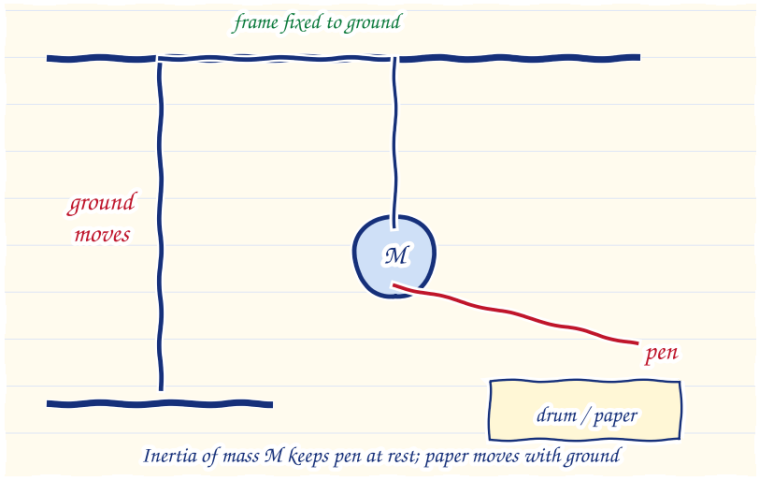


Fig: Principle of a seismograph: heavy mass stays still, frame moves

- Seismoscope - simple device that only indicates that a shock occurred (and maybe its direction/rough amplitude); no time scale, no true record.
- Seismograph - records ground motion vs time with a mass-spring-damper (inertial) sensor, amplifier and recorder; types: short-period, long-period, broadband; horizontal and vertical components. Natural period of sensor is selected so that ground displacement ($T_n >> T$) or ground acceleration ($T_n << T$) is recorded.
- Seismogram - the record produced (trace of ground displacement/velocity/acceleration vs time). P, S and surface wave arrivals are visible; S-P interval -> epicentral distance; amplitude -> magnitude.
- Accelerograph (strong-motion instrument) - records ground acceleration at strong shaking without going off-scale (SMA-1 in India); the record is an accelerogram giving PGA, frequency content, duration. Three components (2 horizontal + 1 vertical).
- Modern: digital broadband sensors, GPS, networks of the National Seismological Network (NCS, IMD); epicentre located by triangulation from at least 3 stations.

1.8 Strong Ground Motion

Strong ground motion is the shaking strong enough to affect structures. Its engineering characteristics:

- Amplitude: Peak Ground Acceleration (PGA), velocity (PGV) and displacement (PGD). Zone V design PGA ~ 0.18 g (DBE) / 0.36 g (MCE).
- Frequency content: dominant periods (response spectrum / Fourier spectrum). Hard rock -> high frequency; soft soil -> long periods, amplified.
- Duration: bracketed or significant duration - longer shaking = more damage accumulation (low-cycle fatigue).
- Factors: magnitude, source mechanism, distance, path (attenuation), local soil conditions (site amplification, liquefaction), directivity, near-fault (fling/pulse) effects.
- Typical accelerograms: 1940 El Centro (PGA 0.32 g), 1995 Kobe, 2001 Bhuj.

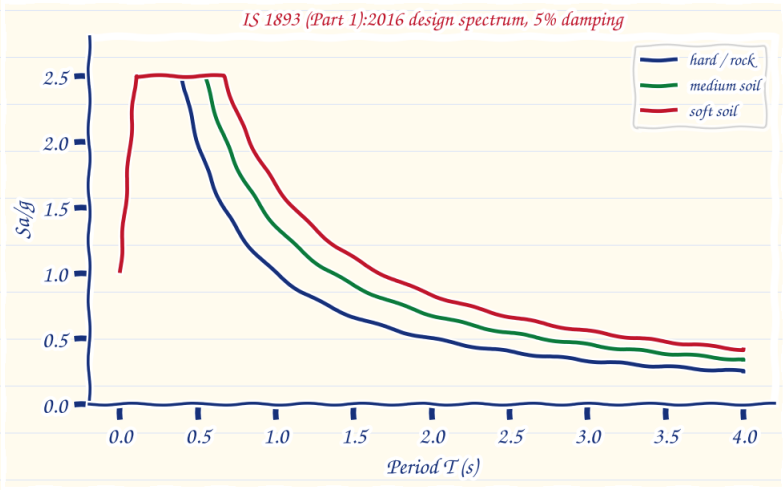


Fig: IS 1893 (Part 1):2016 spectra for different soils

1.9 Seismic Zones of India (IS 1893 (Part 1):2016)

The 2016 revision merges the old Zone I with Zone II -> only four zones: II, III, IV and V (earlier code 2002 had Zone I-V; old Zone I removed). Map is based on maximum considered earthquake (MCE) intensity and past seismicity.

Zone	Seismic intensity	Zone factor Z (MCE)	Design Z/2 (DBE)	Areas (examples)
II	Low (MSK VI)	0.10	0.05	Parts of south India: Bengaluru-Tamil Nadu interior, Rajasthan west
III	Moderate (VII)	0.16	0.08	Chennai, Mumbai, Kolkata, Hyderabad, Kerala, much of A.P.
IV	Severe (VIII)	0.24	0.12	Delhi, Patna, Srinagar region, Sikkim, parts of Gujarat & Bihar
V	Very severe (IX +)	0.36	0.18	Kashmir, Himachal W., Uttarakhand, N-E states, Rann of Kutch, Andaman & Nicobar

Important / Exam point

Remember Z values 0.10, 0.16, 0.24, 0.36 -> Z/2 is the DBE (design basis earthquake) factor used in A_h. About 59% of India's land area is prone to moderate-severe shaking; ~11% falls in Zone V.

- *Zone factor Z = effective peak ground acceleration (as fraction of g) for MCE on rock/firm ground.*
- *Seismic microzonation (soil, liquefaction, slope) refines the national map for cities.*

PART B : THEORY OF VIBRATIONS

2.1 Elements of a Vibratory System

- *Mass (m) - stores kinetic energy; inertia force = m ü.*

- *Spring / stiffness (k)* - stores strain energy; restoring force = $k\,u$. For a cantilever column $k = 3EI/h^3$; for a fixed-fixed column $k = 12EI/h^3$.
- *Damper (c)* - dissipates energy; viscous damping force = $c\,\dot{u}$ (proportional to velocity).
- *Excitation* - external force $p(t)$ or ground motion $\ddot{u}_g(t)$.

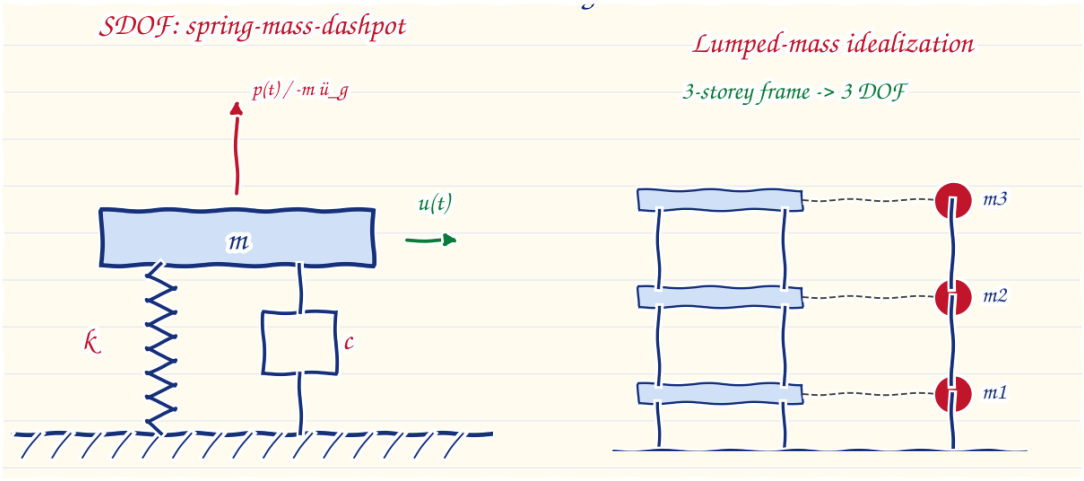


Fig: SDOF model and lumped mass idealization

2.2 Degrees of Freedom and Idealization

> **Definition - Degree of Freedom (DOF)**
> The *minimum number of independent coordinates* needed to completely define the position of all masses of a system at any instant.

- *SDOF* - one coordinate (water tank on a single column, single-storey frame with rigid girder).
- *MDOF* - n coordinates for an n-storey shear building (n lumped masses, each with one lateral translation) -> n natural frequencies.
- *Continuous system* - mass and elasticity distributed (beams, chimneys, dams); *infinite DOF*, governed by partial differential equations.
- *Lumped mass idealization* - replace distributed mass by discrete masses at floor levels (floor mass includes slab, beams, half of columns above and below). Assumptions: floor slabs rigid in-plane, columns massless & axially rigid -> *shear building*; it reduces a continuous problem to a finite number of DOF.

Equation of motion for an n-DOF shear building: $[M]\{\ddot{u}\} + [C]\{\dot{u}\} + [K]\{u\} = -[M]\{1\}\ddot{u}_g$ (K is tridiagonal).

2.3 Oscillatory Motion and Simple Harmonic Motion

- *Periodic motion* repeats after equal time intervals; *harmonic* motion is sinusoidal: $u = U \sin(\omega t + \varphi)$.
- *Simple Harmonic Motion (SHM)*: acceleration proportional to displacement and directed to the mean position: $\ddot{u} = -\omega^2 u$.
- Amplitude U, circular frequency ω (rad/s), cyclic frequency $f = \omega/2\pi$ (Hz), period $T = 1/f = 2\pi/\omega$ (s), phase angle φ.

- Velocity leads displacement by 90°, acceleration by 180° (so max velocity = ωU, max acceleration = $\omega^2 U$).
- Complex form: $u = Re[U e^{i\omega t}]$; two harmonics of same frequency add as vectors (phasors).

2.4 Free Vibration of an SDOF System

(a) Undamped

Equation of motion (Newton / D'Alembert)

$m\ddot{u} + ku = 0$

$\omega_n = \sqrt{(k/m)}$ (rad/s), $T_n = 2\pi/\omega_n = 2\pi\sqrt{(m/k)}$, $f_n = 1/T_n$

Solution: $u(t) = u_0 \cos \omega_n t + (\dot{u}_0/\omega_n) \sin \omega_n t$ (constant amplitude, no decay)

Amplitude $U = \sqrt{[u_0^2 + (\dot{u}_0/\omega_n)^2]}$

Natural period depends only on m and k (not on the initial conditions). Stiffer/lighter -> shorter period. Rough building period: $T \approx 0.1\,N$ seconds (N = number of storeys).

(b) Damped (viscous)

Equation of motion

$m\ddot{u} + c\dot{u} + ku = 0 \rightarrow \ddot{u} + 2\zeta\omega_n \dot{u} + \omega_n^2 u = 0$

Critical damping $c_{cr} = 2\sqrt{(km)} = 2m\omega_n$ Damping ratio $\zeta = c / c_{cr}$

Characteristic roots: $s = -\zeta\omega_n \pm \omega_n \sqrt{(\zeta^2 - 1)}$

Case	Condition	Behaviour
Under-damped	$\zeta < 1$ (all real buildings)	Decaying oscillation: $u = e^{-\zeta\omega_n t}[u_0 \cos \omega_d t + ((\dot{u}_0 + \zeta\omega_n u_0)/\omega_d) \sin \omega_d t]$, damped frequency $\omega_d = \omega_n \sqrt{(1-\zeta^2)}$, $T_d = 2\pi/\omega_d$
Critically damped	$\zeta = 1$ ($c = c_{cr}$)	No oscillation; returns to rest in minimum time: $u = (A + Bt) e^{-\omega_n t}$
Over-damped	$\zeta > 1$	Sluggish non-oscillatory return, slower than critical

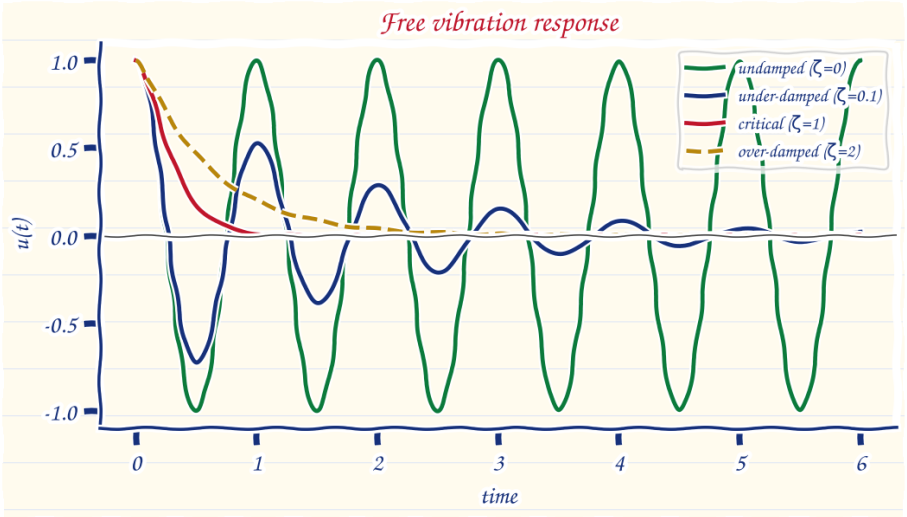


Fig: Free vibration for different damping ratios

- Typical ζ: steel buildings 2-5%, RC buildings 5% (IS 1893 spectra are for 5% damping), masonry 5-7%, bridges 2-5%, structure with dampers 10-30%.
- For $\zeta \le 20\%$, $\omega_d \approx \omega_n$ (difference < 2%). Damping reduces amplitude, not period (practically).

2.5 Logarithmic Decrement

Definition - Logarithmic decrement (δ)
Natural logarithm of the ratio of any two successive peak amplitudes in free damped vibration. It measures the damping experimentally.

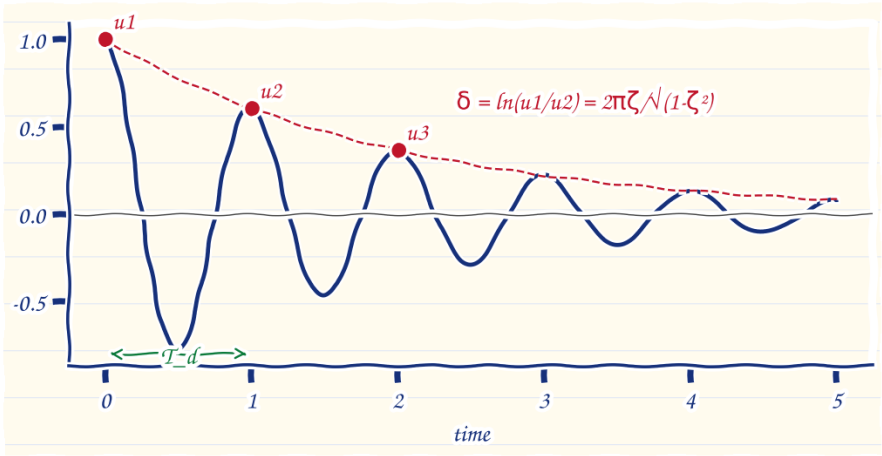


Fig: Decay of free vibration - peaks u1, u2, u3

Formulae
$\delta = \ln(u_1/u_2) = \zeta \omega_n T_d = 2\pi\zeta / \sqrt{(1 - \zeta^2)} \approx 2\pi\zeta$ (for small ζ)
For n cycles apart: $\delta = (1/n) \ln(u_1/u_{n+1})$ -> $\zeta \approx \delta/2\pi$
Amplitude after n cycles: $u_{n+1} = u_1 e^{-n\delta}$

Important / Exam point
Numerical (typical): Peak amplitude falls from 20 mm to 5 mm in 5 cycles -> $\delta = (1/5) \ln(20/5) = 0.277$ -> $\zeta = 0.277/2\pi = 0.044 \approx 4.4\%$.

2.6 Forced Vibration - Harmonic Excitation

Equation: $m\ddot{u} + c\dot{u} + ku = p_0 \sin \omega t$
Total response = complementary (transient, dies out due to damping) + particular (steady state).
Steady state: $u(t) = \rho \sin(\omega t - \varphi)$, static deflection $u_{st} = p_0/k$
$\rho = u_{st} \cdot D$, $D = 1 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, $r = \omega/\omega_n$
Phase lag $\varphi = \tan^{-1}[2\zeta r / (1 - r^2)]$

Definition - Dynamic Magnification Factor (DMF / D)
Ratio of the dynamic amplitude to the static deflection under the same force amplitude: $D = \rho/u_{st}$.

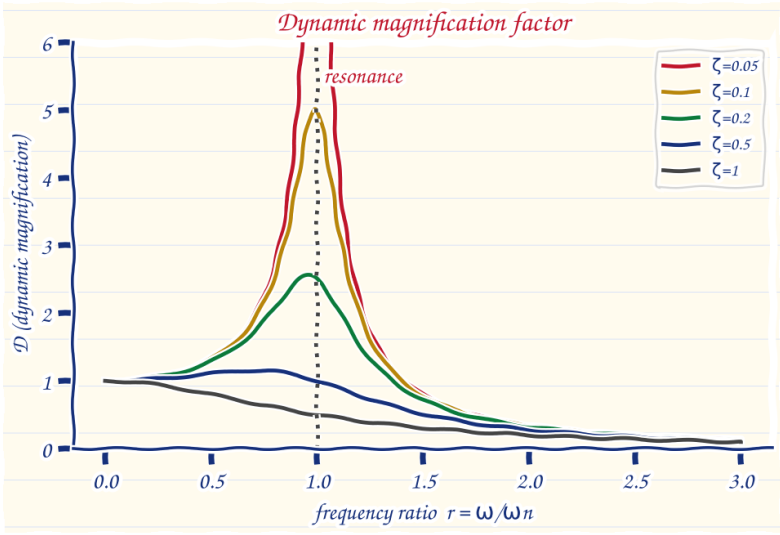


Fig: DMF vs frequency ratio

- $r << 1$ (slow load): $D \approx 1$ - behaves statically (stiffness controlled).
- $r = 1$ (resonance): $D = 1/2\zeta$ ($max_r \approx 1/2\zeta$; infinite if undamped) - damping controlled; $\varphi = 90°$. For $\zeta = 5\%$ $D = 10$!
- $r >> 1$ (fast load): $D \to 0$ - mass (inertia) controlled; structure hardly responds; basis of base isolation.
- Peak of D occurs at $r = \sqrt{(1 - 2\zeta^2)}$ with $D_{max} = 1/(2\zeta\sqrt{(1 - \zeta^2)})$.
- Resonance danger: when ground motion's dominant period ≈ building period (Mexico City 1985: soft lake bed $T \approx 2$ s vs 6-15 storey buildings).

Transmissibility and force transmitted

Force transmitted to support / isolation: $TR = \sqrt{[1 + (2\zeta r)^2]} / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$; isolation effective only for $r > \sqrt{2}$.

2.7 Excitation by Rigid Base Translation (Earthquake)

For ground displacement $u_g(t)$, with u = relative displacement of mass w.r.t. base, total displacement $u_t = u + u_g$:

Equation of motion for ground motion

$m\ddot{u} + c\dot{u} + ku = -m\ddot{u}_g(t) \to \ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2 u = -\ddot{u}_g(t)$

Effective earthquake force $p_{eff}(t) = -m\ddot{u}_g(t)$ (inertia force from base acceleration)

For harmonic base motion $\ddot{u}_g = a \sin \omega t$: relative amplitude $u_0 = (m a/k) D = (a/\omega_n^2) D$

- Only ground acceleration and system properties (ω_n, ζ) control relative response; mass cancels out: response depends on T_n, ζ only.
- Shear force in the spring (column) $V = ku = m\omega_n^2 u \to$ pseudo-acceleration $A = \omega_n^2 u$ gives equivalent static force $f_s = mA$.

2.8 Earthquake Ground Motion - Response to General Ground Acceleration

Duhamel integral (damped SDOF)

$u(t) = -(1/\omega_d) \int_0^t \ddot{u}_g(\tau)\, e^{-\zeta\omega_n(t-\tau)} \sin \omega_d(t-\tau)\, d\tau$

- The accelerogram is applied to many SDOF oscillators with different T_n (same ζ); the maximum response is plotted against T_n -> response spectrum: $S_d = \max|u|$, $S_v = \omega_n S_d$ (pseudo-velocity), $S_a = \omega_n^2 S_d$ (pseudo-acceleration).
- Design Spectrum (smoothed, IS 1893 S_d/g vs T) is the envelope/mean of many spectra for a soil class - used for code analysis.
- Base shear for SDOF: $V_b = m S_a = (S_d/g) W$.
- Earthquake ground motion is random, transient, broad-band with 3 translational components; vertical usually 2/3 of horizontal.

SOLVED EXAMPLES (typical exam numericals)

Ex.1 Natural period of SDOF

Water tank: $m = 20\,000$ kg on column of lateral stiffness $k = 8 \times 10^6$ N/m.

$\omega_n = \sqrt{(k/m)} = \sqrt{(8e6/20000)} = \sqrt{400} = 20$ rad/s

$T_n = 2\pi/\omega_n = 0.314$ s, $f_n = 1/T_n = 3.18$ Hz

Ex.2 Damping ratio and log decrement

Same tank has dashpot $c = 40\,000$ N.s/m.

$c_{cr} = 2\sqrt{(km)} = 2\sqrt{(8e6 \times 20000)} = 8 \times 10^5$ N.s/m -> $\zeta = 40000/800000 = 0.05$ (5%)

$\omega_d = 20\sqrt{(1 - 0.05^2)} = 19.97$ rad/s (almost $= \omega_n$)

$\delta = 2\pi(0.05)/\sqrt{(1-0.05^2)} = 0.3145$ -> $u_1/u_2 = e^{0.3145} = 1.37$ (each peak is 73% of the previous one)

Ex.3 Dynamic magnification

$\zeta = 0.05$, $r = \omega/\omega_n = 0.8$: $D = 1/\sqrt{[(1-0.64)^2 + (2 \times 0.05 \times 0.8)^2]} = 1/\sqrt{(0.1296 + 0.0064)} = 2.71$

At resonance $r = 1$: $D = 1/(2\zeta) = 10$ -> a harmonic force of 10 kN produces 10 times the static deflection (10/k).

Ex.4 Energy and epicentral distance

M 6 -> M 8: $E_8/E_6 = 10^{1.5 \times 2} = 1000$ times.

S-P lag $\Delta t = 20$ s, $V_p = 8$ km/s, $V_s = 4.6$ km/s: $d = \Delta t \times V_p V_s/(V_p - V_s) = 20 \times (8 \times 4.6)/3.4 =$ ~216 km.

Formula sheet

Short-answer table: $\omega_n = \sqrt{(k/m)}$; $c_{cr} = 2\sqrt{(km)}$; $\zeta = c/c_{cr}$; $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$; $\delta \approx 2\pi\zeta$; $D_{res} = 1/2\zeta$; $Sa = \omega_n^2 Sd$; $V_b = (Sa/g)W$.

QUICK REVISION - Likely questions

1. Explain elastic rebound theory and types of faults with sketches.

2. Describe plate tectonics and the types of plate boundaries; why does India experience earthquakes?
3. Classify seismic waves; derive/state V_p and V_s; sequence of arrival.
4. Differentiate magnitude and intensity; Richter scale, energy relation $\log E = 1.5M + 4.8$.
5. Seismograph vs seismoscope vs accelerograph; characteristics of strong ground motion.
6. Seismic zones of India as per IS 1893 (Part 1):2016 with Z values.
7. Derive free vibration response of damped SDOF; define critical damping; log decrement.
8. Define DMF; derive expression; effect of damping near resonance.
9. Derive equation of motion of SDOF subjected to ground motion.